Piecewise-Defined & Implicit Functions

Evaluating piecewise functions

1 Given $f(x) = x^2$ for $x < 0$, and $f(x) = 2x + 1$ for $x \geq 0$, find $f(-3)$ and $f(4)$.

2 A shipping company charges a flat 5 dollars for packages up to 2 kg, plus 1.50 dollars per kg for anything over 2 kg. Write a piecewise function $C(w)$ for the cost, and find $C(5)$.

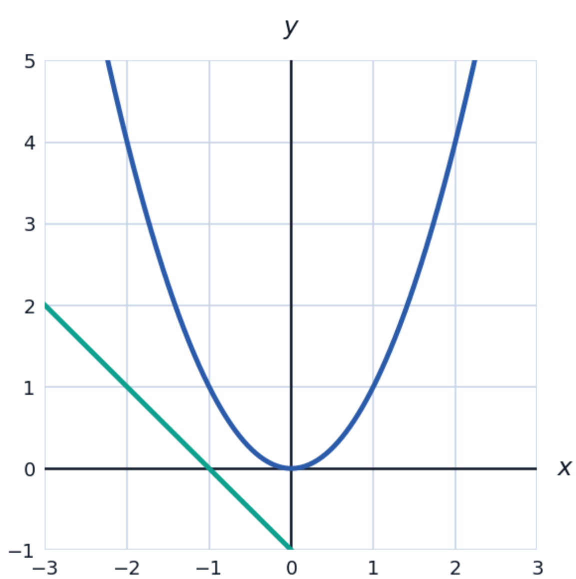
Checking continuity of piecewise functions

3 Find the value of k that makes $f(x) = 3x + k$ for $x < 2$, and $f(x) = x^2 - 1$ for $x \geq 2$, continuous at $x = 2$.

4 Determine whether $g(x) = x + 1$ for $x \leq 1$, and $g(x) = 3x - 2$ for $x > 1$, is continuous at $x = 1$, showing your reasoning.

Visualizing a piecewise graph

5 The graph shows $f(x) = -x - 1$ for $x < 0$, and $f(x) = x^2$ for $x \geq 0$.



a Evaluate the left piece and the right piece of f at $x = 0$, and use this to determine whether f is continuous there.

b State the range of f shown on the graph.

Implicit relations and testing for functions

6 Determine whether $x^2 + y^2 = 25$ defines y as a function of x. Justify using the vertical line test.

7 Solve $x^2 + y^2 = 25$ for y explicitly, and state which branch corresponds to a function.

Implicit differentiation-style reasoning (algebraic slope)

8 For the implicit relation $x^2 + 4y^2 = 16$ (an ellipse), find the two y-values when $x = 2$, and describe what this means about the relation as a function.

Modeling with piecewise functions

9 A tax schedule charges 10% on income up to 10{,}000 dollars and 20% on income above 10{,}000 dollars. Write a piecewise tax function $T(x)$ for income x, and find the tax owed on an income of 25{,}000 dollars.

Piecewise functions with three or more pieces

10 A parking garage charges 3 dollars for the first hour, 2 dollars for each additional hour up to 5 hours total, and a flat 15 dollars for anything over 5 hours. Write this as a three-piece function $P(h)$ for h hours parked, and find $P(4)$.

11 Evaluate the absolute value function $f(x) = |2x - 6|$ at $x = 1$ and $x = 5$ by first rewriting it as a two-piece function without absolute value bars.

Graphing implicit relations point-by-point

12 For the implicit relation $y^2 = x + 4$, find all y-values when $x = 0$, $x = 5$, and $x = -4$, and state the smallest allowed x-value.

Piecewise functions and average rate of change

13 For $f(x) = x^2$ on $x < 2$ and $f(x) = 3x - 2$ on $x \geq 2$, find the average rate of change of f on $[0, 4]$.

14 Given $g(x) = 2x + 1$ for $x \leq 3$ and $g(x) = x^2 - 5$ for $x > 3$, find the average rate of change of g on $[1, 5]$.

15 A cell phone plan charges 20 dollars flat for up to 2 GB of data, plus 8 dollars per GB for each additional GB, up to a maximum monthly charge of 68 dollars (unlimited data cap). Write a three-piece function $M(x)$ for the monthly cost based on x GB used, and find $M(8)$.